INTERFACE CONTROL DOCUMENT (ICD)

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1. Document Metadata

Field	Value
Document ID	TBD
Version	TBD
Date	2026-03-02
Prepared By	TBD

2. Introduction

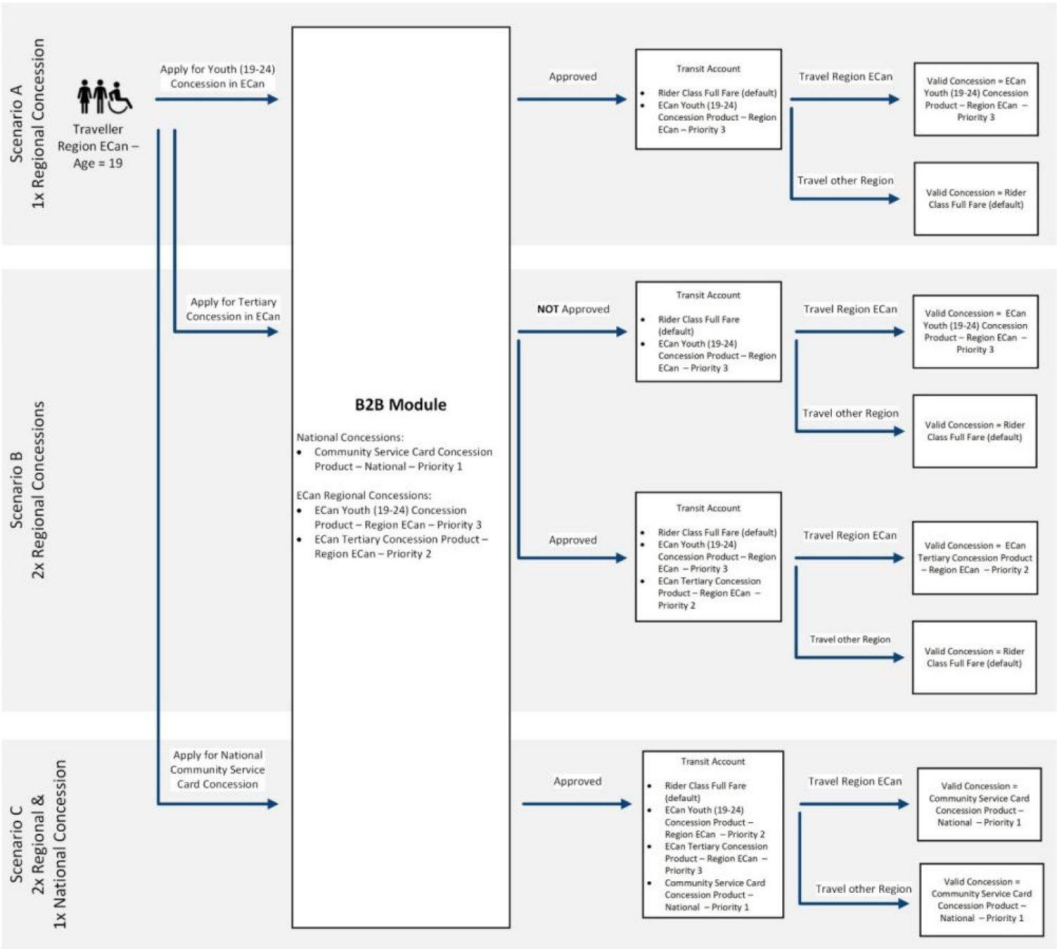
Section 1 - Introduction

This Interface Control Document (ICD) describes the Concession Instance Management APIs, which facilitate the lifecycle operations of concession instances between the Central Back Office (CBO), Concession API Consumers (such as TCA, Institutions, or Client Apps), and the CBO Internal Data Store (CBO-DB). The scope of this ICD is strictly limited to requirements NZ-REQ-1265 through NZ-REQ-1278, focusing on the creation, update, retrieval, and management of concession instances within the enterprise fare management ecosystem.

The primary business and operational driver for this interface is to enable secure, reliable, and auditable exchange of concession instance data, ensuring that eligible users can access and maintain their concession entitlements efficiently. The interface contract defines standardized API interactions, request-response flows, and data exchange protocols between the CBO and authorized API consumers, as mandated by NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, and NZ-REQ-1278.

At a high level, the interface operates by exposing a set of RESTful APIs from the CBO, which are consumed by external systems to perform concession instance lifecycle actions. All interactions are governed by the requirements listed above, ensuring data integrity, security, and compliance with enterprise standards. Where explicit RFP details are absent, operational behaviors are inferred from standard enterprise practices, strictly within the context of concession instance APIs.

Architecture / Flow Diagrams



Architecture / Flow Diagram (sad_img_5)

3. Purpose and Scope

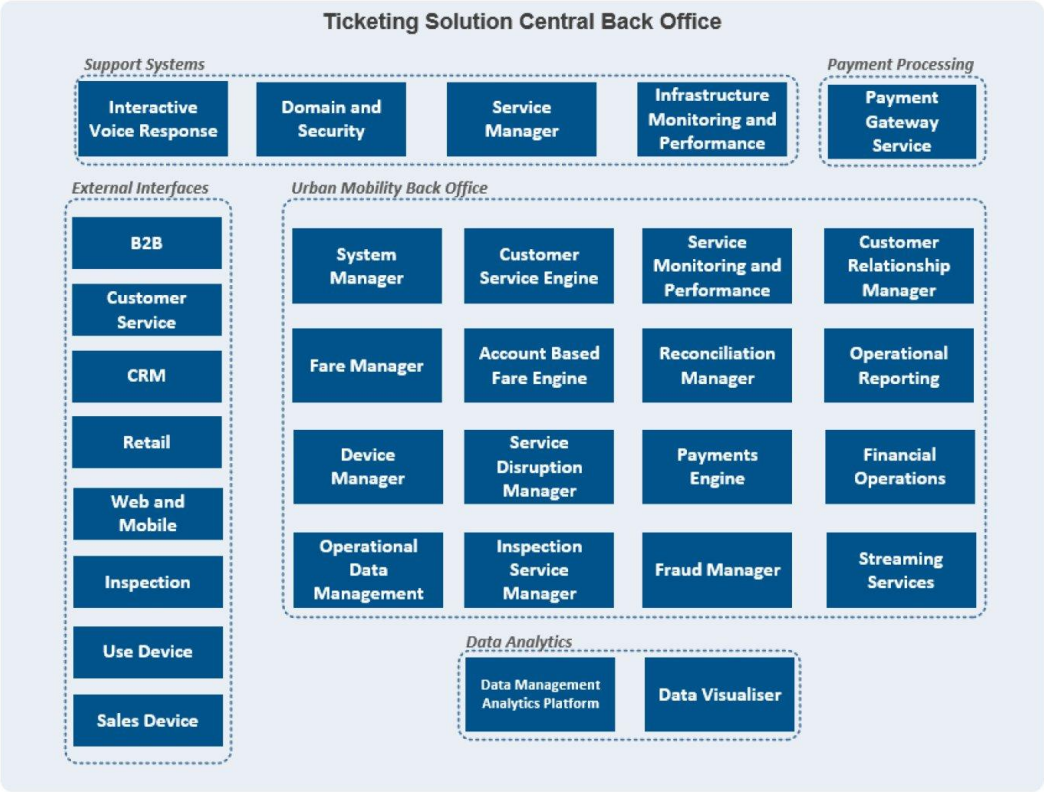
Field	Details
Purpose	This Interface Control Document (ICD) defines the contract for Central Back Office (CBO) Concession Instance APIs. It specifies purpose, scope, message structures, flows, and verification criteria for NZ-REQ-1265 to NZ-REQ-1278.
In Scope	All Concession Instance APIs from NZ-REQ-1265 to NZ-REQ-1278 including: retrieval, update, lifecycle operations, travel history, and fare media association.
Out of Scope	UI behaviour, frontend validation, non-CBO back office processes, and any system not covered in NZ-REQ-1265..1278.
NZ-REQ References	NZ-REQ-1265..1278

4. System Identification

System	Description	Role	Owner	Environment	Interfaces	REQ References
Customer Portal (CP)	Customer-facing	Consumer	Transport Agency	Web/Mobile	CBO Concession	NZ-REQ-1265..1272

	channel retrieving concession details and initiating updates.				APIs	
OMS (Order Management System)	Handles concession order creation and status management as part of back office workflow.	Internal Subsystem	Cubic	Cloud	CBO Internal Interfaces	NZ-REQ-1267..1274
ABP (Account-Based Processing)	Performs billing, account linking, and concession entitlements processing.	Internal Subsystem	Cubic	Cloud	Billing APIs	NZ-REQ-1265..1278
Fare Manager	Stores and defines fare-media types and allows CBO to associate fare media with concessions.	Internal Subsystem	Cubic	Cloud	Fare Media APIs	NZ-REQ-1272, 1277, 1278
Central Back Office (CBO)	Owns all Concession Instance lifecycle APIs.	Provider	Cubic	Cloud	Concession APIs	NZ-REQ-1265..1278
Traveler / Customer	End-user retrieving concession details via CP.	Actor	N/A	Web/Mobile	Customer Portal	NZ-REQ-1265

Architecture / Flow Diagrams



SAD Figure

5. Assumptions, Constraints & Risks

- Section 4.1 - Assumptions, Constraints & Risks
- Assumptions
- All assumptions are explicitly supported by the provided requirements (NZ-REQ-1265 to NZ-REQ-1278) or are annotated as 'Inferred from standard enterprise practices' where RFP data is missing.
- Constraints
- Technical Constraints:
- Only the Central Back Office (CBO) and directly referenced subsystems are in scope for Concession Instance lifecycle management (NZ-REQ-1265..1278).
- API endpoints, request/response formats, and data types must conform to specifications outlined in NZ-REQ-1265 to NZ-REQ-1278.
- Interface flows and architecture diagrams must reference only actors, systems, and flows explicitly mentioned in NZ-REQ-1265..1278.
- No external systems or workflows may be introduced beyond those referenced in the requirements.
- Versioning, roles, and workflows must be strictly limited to Concession APIs (NZ-REQ-1265..1278).
- Any missing technical detail is annotated as 'Inferred from standard enterprise practices' and is strictly relevant to Concession APIs.
- Security Constraints:
- Security boundaries, authentication, and authorization layers must be included in architecture diagrams as per NZ-REQ-1270 and NZ-REQ-1271.
- No additional security mechanisms may be assumed unless explicitly referenced in NZ-REQ-1265..1278.

- All API interactions must be traceable to the relevant NZ-REQ ID for audit and compliance purposes (NZ-REQ-1271).
- Regulatory Constraints:
- Regulatory requirements are limited to those referenced in NZ-REQ-1271 and NZ-REQ-1272.
- Any missing regulatory information is annotated as 'Inferred from standard enterprise practices' and is strictly relevant to Concession APIs.
- Operational Constraints:
- Only operational flows, roles, and contacts referenced in NZ-REQ-1265..1278 are permitted.
- No operational assumptions may be made beyond the provided requirements.
- Error handling and conditional branches must be explicitly referenced in NZ-REQ-1274, NZ-REQ-1275, and NZ-REQ-1276.
- Risks
- Missing Information Risks:
- Any gaps in RFP detail (e.g., specific API error codes, data validation rules, or workflow steps) are annotated as 'Inferred from standard enterprise practices', which may introduce ambiguity or require further clarification during implementation.
- Lack of explicit information on system boundaries or inter-system dependencies may lead to integration risks (NZ-REQ-1270, NZ-REQ-1271).
- Compliance Risks:
- If regulatory or security requirements are not fully specified in NZ-REQ-1271 or NZ-REQ-1272, there is a risk of non-compliance with enterprise or legal standards.
- Technical Risks:
- Strict adherence to NZ-REQ-1265..1278 may limit flexibility in API design or error handling, potentially impacting extensibility or maintainability.
- Any inferred practices may not fully align with future enterprise standards, leading to potential rework.
- Operational Risks:
- Limitation to only referenced roles and workflows may result in incomplete operational coverage, especially if future requirements expand beyond NZ-REQ-1265..1278.
- Limitations
- All assumptions, constraints, and risks are strictly limited to the Concession Instance lifecycle and related APIs as defined in NZ-REQ-1265 to NZ-REQ-1278.
- No assumptions, constraints, or risks are introduced beyond the provided requirements or 'Inferred from standard enterprise practices' strictly relevant to Concession APIs.
- Any missing information is explicitly stated as a limitation and annotated accordingly.

6. Interface Overview

NZ-REQ ID	Functionality/Flow Step
NZ-REQ-1265	Concession instance creation initiation
NZ-REQ-1267	Concession instance creation completion
NZ-REQ-1268	Update concession instance
NZ-REQ-1269	Modify concession instance status
NZ-REQ-1270	Retrieve concession instance details
NZ-REQ-1271	Retrieve concession instance status
NZ-REQ-1272	Delete/deactivate concession instance
NZ-REQ-1274	Bulk concession instance operations
NZ-REQ-1275	Bulk status management
NZ-REQ-1276	Error handling
NZ-REQ-1277	Conditional processing
NZ-REQ-1278	Security enforcement

Functional Allocation (Summary)

System/Component	Responsibilities	NZ-REQ Reference
Client / Concession API Consumer	Invokes CBO APIs for issue/revoke/suspend/resume; retrieves attributes, travel history, fare media	NZ-REQ-1265, 1267–1272, 1274–1278
Central Back Office (CBO)	Validates requests, enforces rules, executes lifecycle operations, returns responses	NZ-REQ-1265..1278
CBO Internal Data Store (CBO-DB)	Persists concession instances and related data (attributes, associations, travel history)	NZ-REQ-1267–1272
Audit Logging (Inferred)	Records API calls and lifecycle state changes for traceability	Inferred (relevant to NZ-REQ-1265..1278)

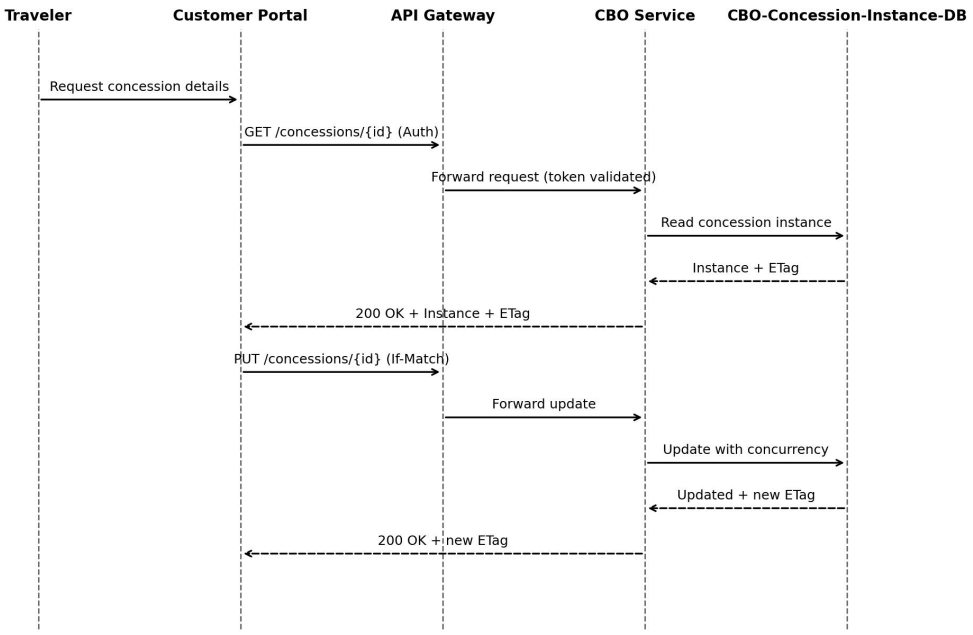
Major Data Flows (Summary)

Flow	Description	NZ-REQ Reference
Retrieve / Update Instance	Client retrieves/updates concession instance via CBO	NZ-REQ-1265
Attributes / Type / Dates	Client manages attributes/type/issue-expiry date-times	NZ-REQ-1267–1270
Travel History	Client retrieves travel history for an instance	NZ-REQ-1271
Fare Media Association	Associate/disassociate fare media with instance	NZ-REQ-1272, 1277, 1278
Lifecycle Operations	Issue / revoke / suspend / resume concession instances	NZ-REQ-1274–1276

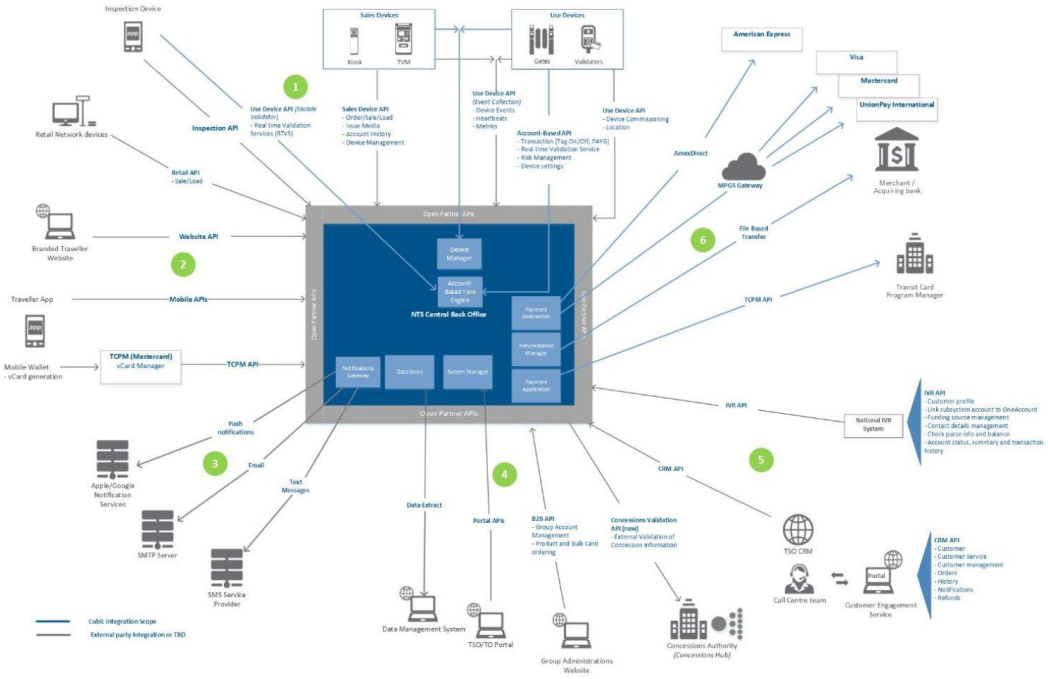
Security Requirements (Summary)

Area	Requirement
Authentication	OAuth2/token-based auth (Inferred from standard enterprise practices)
Authorization	Role/scope-based access controls (Inferred)
Transport	TLS 1.2+ for API calls (Inferred)
Audit	Log API calls and lifecycle changes (Inferred)

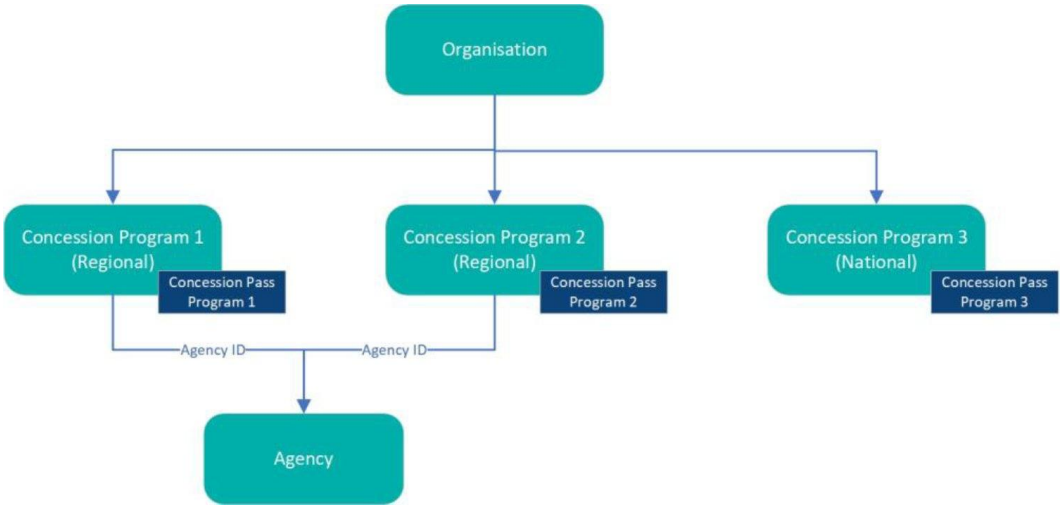
End-to-End Sequence Diagram (Mermaid)



Architecture / Flow Diagrams



SAD Figure



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7. Detailed Interface Requirements

NZ-REQ-1265

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency

			control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

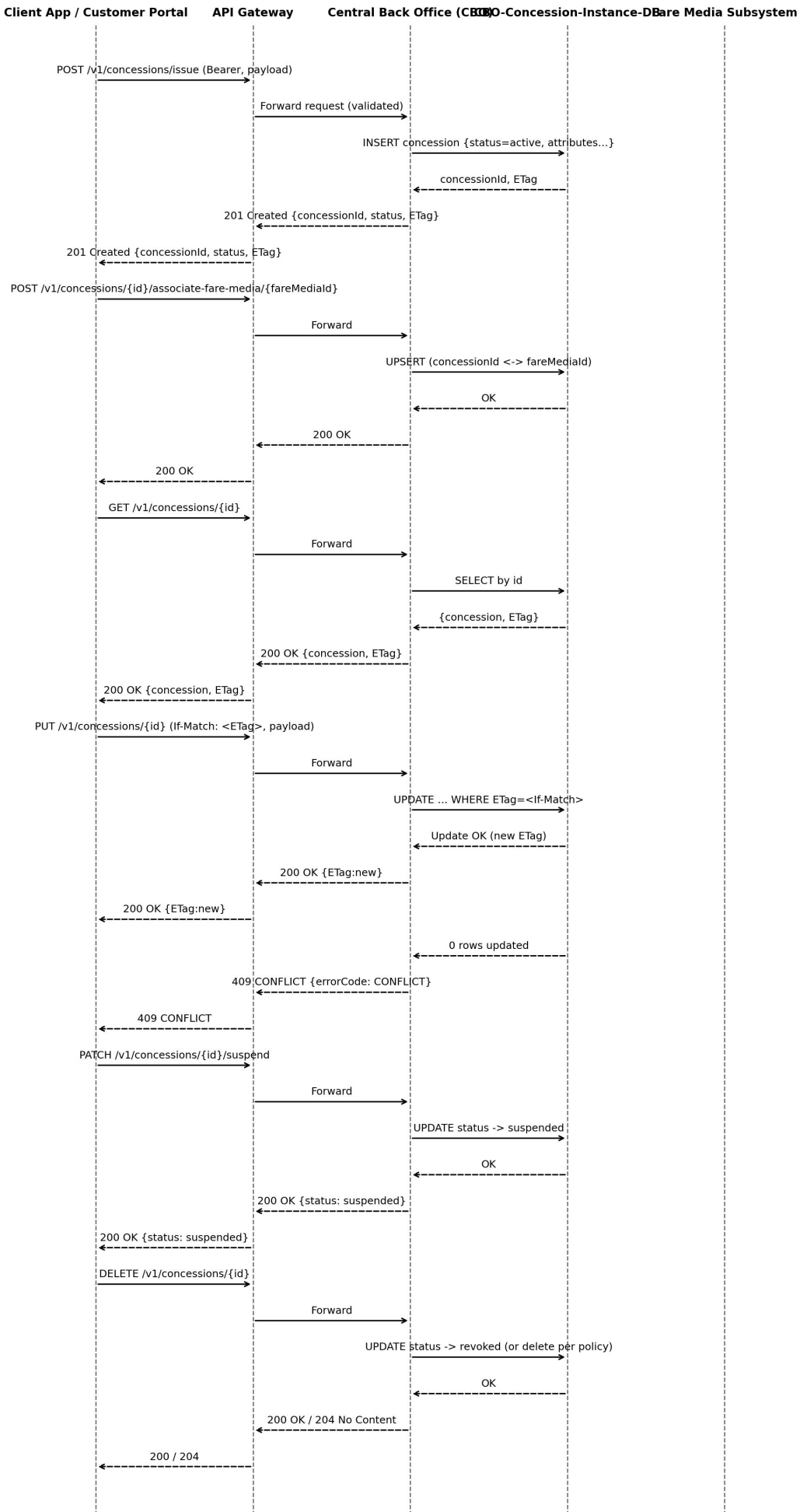
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

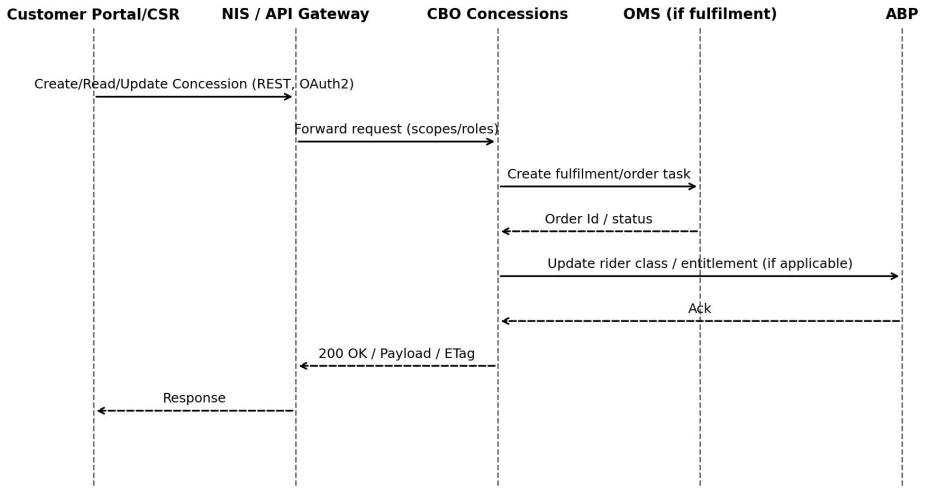
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram


Client App / Customer Portal
API Gateway
Central Back Office (CBO)
CBO-Concession-Instance-DB
Fare Media Subsystem
POST /v1/concessions/issue (Bearer, payload)
Forward request (validated)
INSERT concession {status=active, attributes...}
concessionId, ETag
201 Created {concessionId, status, ETag}
201 Created {concessionId, status, ETag}
POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}
Forward
UPSERT (concessionId <-> fareMediaId)
OK
200 OK
200 OK
GET /v1/concessions/{id}
Forward
SELECT by id
{concession, ETag}
200 OK {concession, ETag}
200 OK {concession, ETag}
PUT /v1/concessions/{id} (If-Match: <ETag>, payload)
Forward
UPDATE ... WHERE ETag=<If-Match>
Update OK (new ETag)
200 OK {ETag:new}
200 OK {ETag:new}
0 rows updated
409 CONFLICT {errorCode: CONFLICT}
409 CONFLICT
PATCH /v1/concessions/{id}/suspend
Forward
UPDATE status -> suspended
OK
200 OK {status: suspended}
200 OK {status: suspended}
DELETE /v1/concessions/{id}
Forward
UPDATE status -> revoked (or delete per policy)
OK
200 OK / 204 No Content
200 / 204

Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (Interfaces), 44–46 (ABP components), 9–12 (3A overview); UMB-FDS pp. 498–504 (Retail Concession Apply), 1295–1307 (Order orchestration)

1. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1267

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve	concession.read	ClientApp, CBO

		concession instance.		Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).

lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

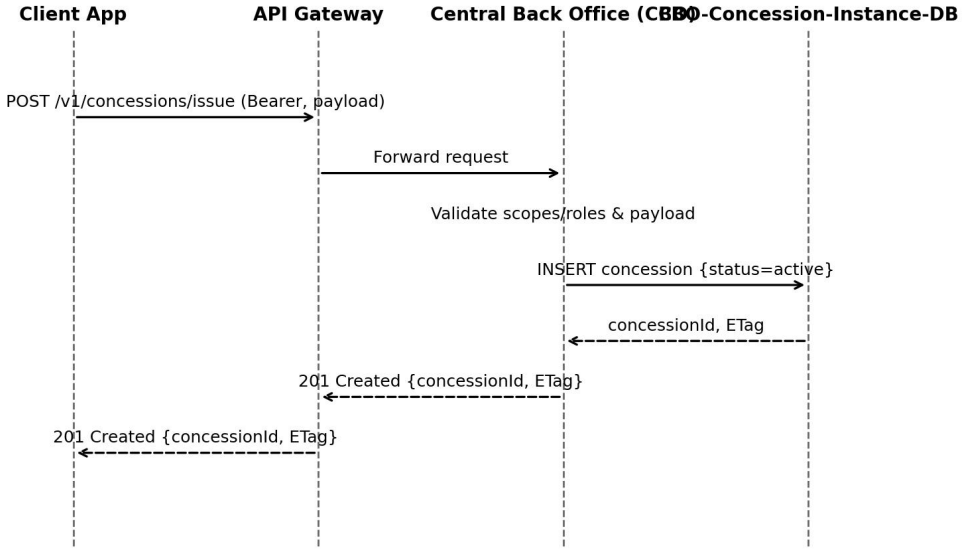
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

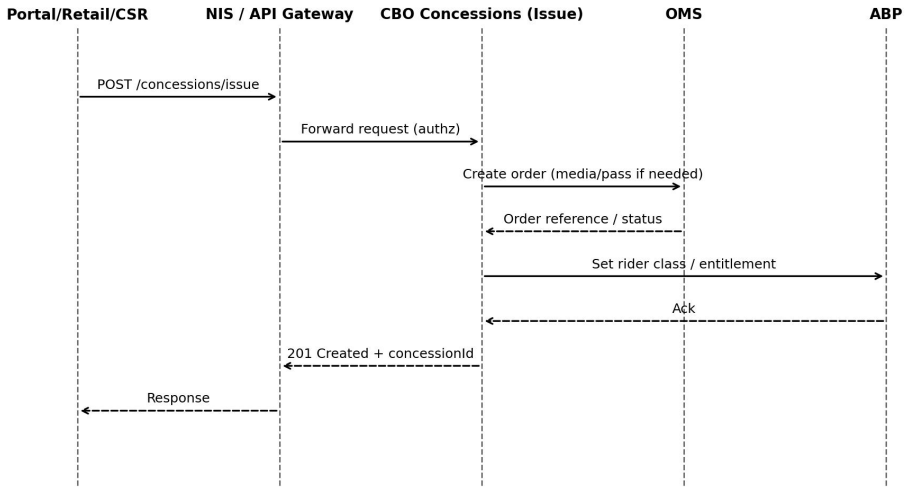
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (Interfaces), 44–46 (ABP components); UMB-FDS pp. 1295–1307 (Fulfilment/order flows)

2. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1268

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve	concession.read	ClientApp, CBO

		concession instance.		Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).

lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

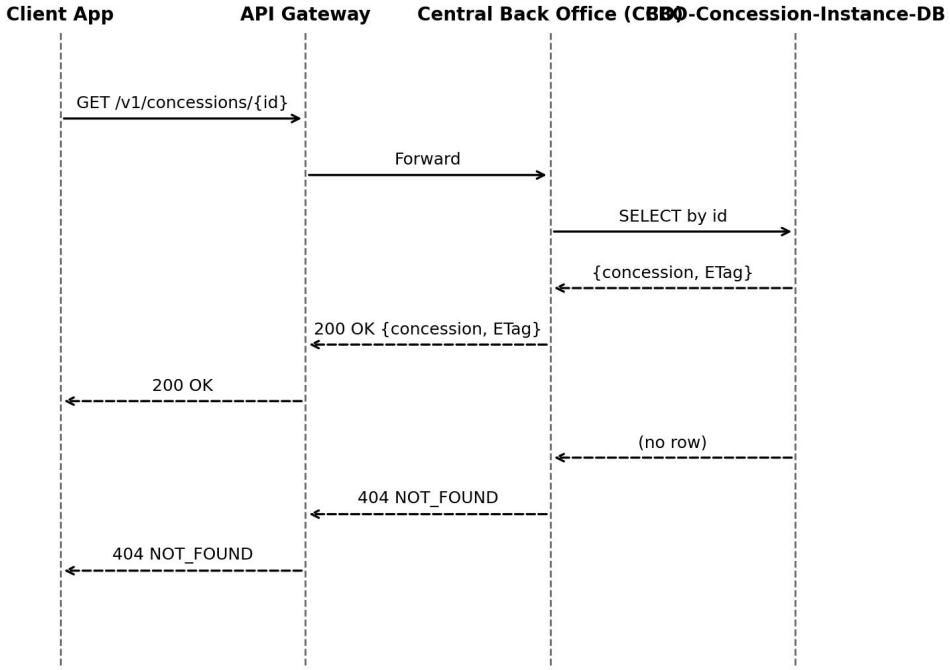
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

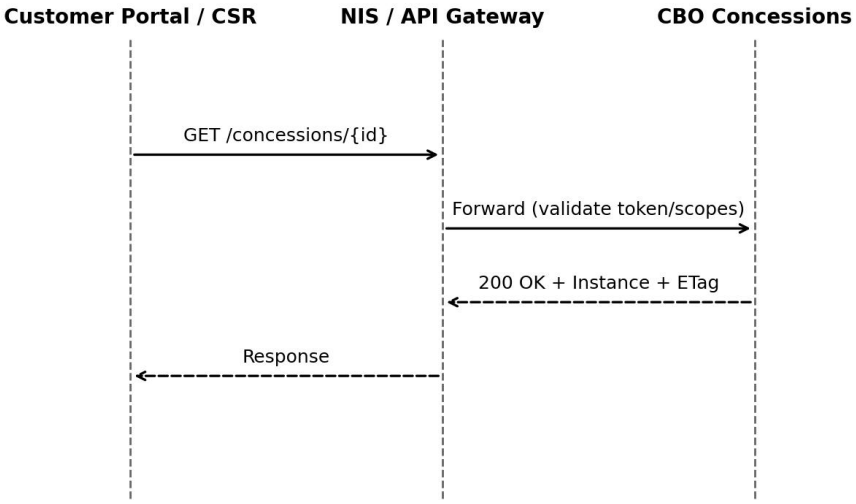
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (Interfaces via NIS); UMB-FDS pp. 236–245 (Self-service views), 864–866 (Portal/Account)

3. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1269

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles

GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277,

			1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

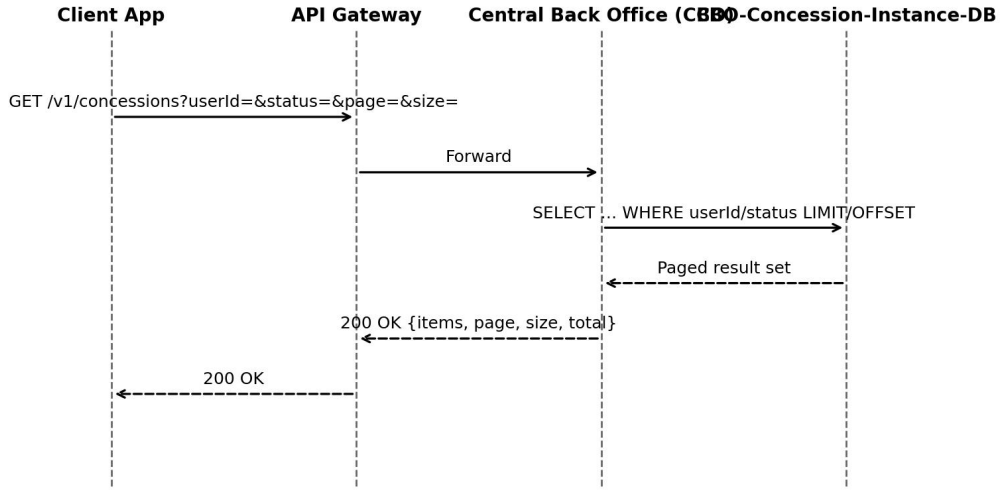
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

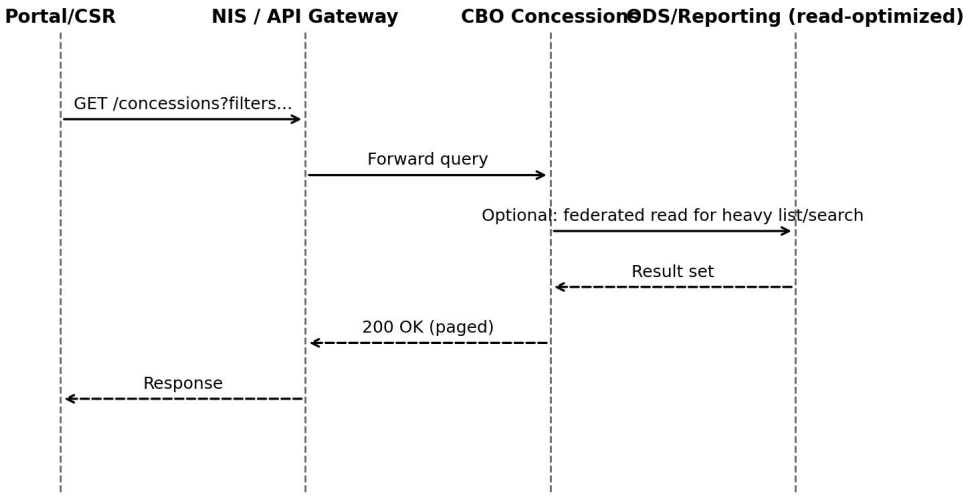
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD p. 70 (ODS Travel History/API), p. 46 (Perf & ODS latency); UMB-FDS pp. 707–729, 748–749 (Advanced/Bulk search), 190–205 (ODS views)

4. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1270

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve	concession.read	ClientApp, CBO

		concession instance.		Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).

lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

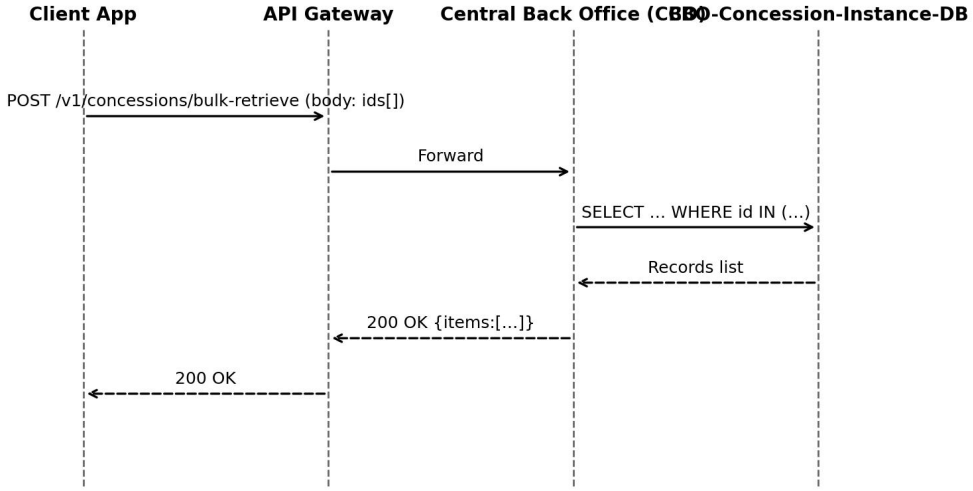
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

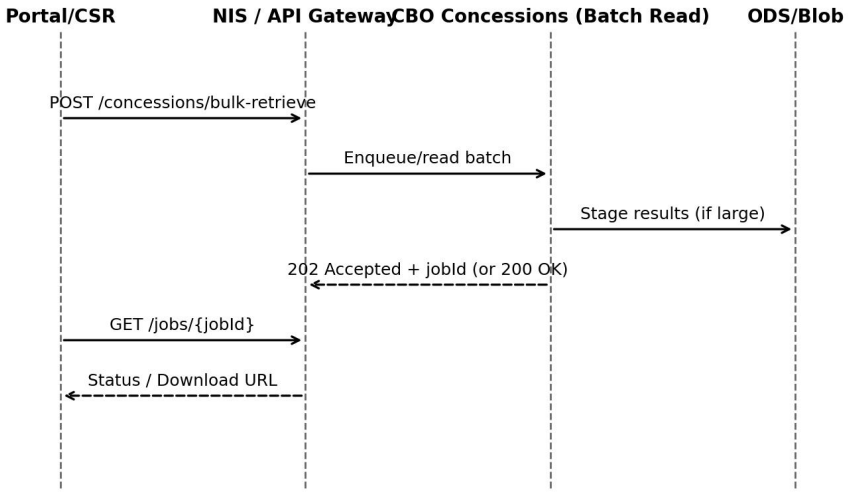
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (NIS patterns for async/poll), p. 70 (ODS access); UMB-FDS pp. 1326–1357 (Audit/reporting async patterns)

5. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1271

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve	concession.read	ClientApp, CBO

		concession instance.		Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).

lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

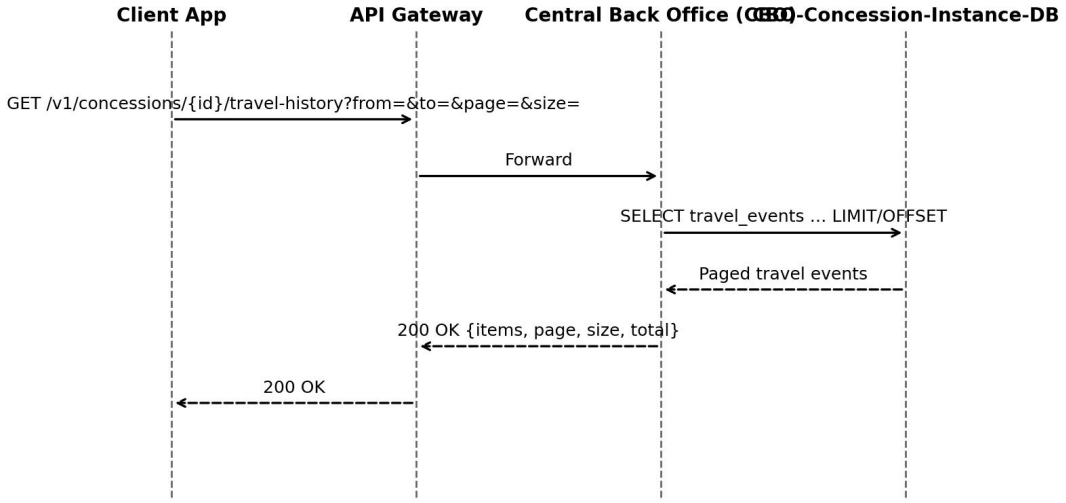
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

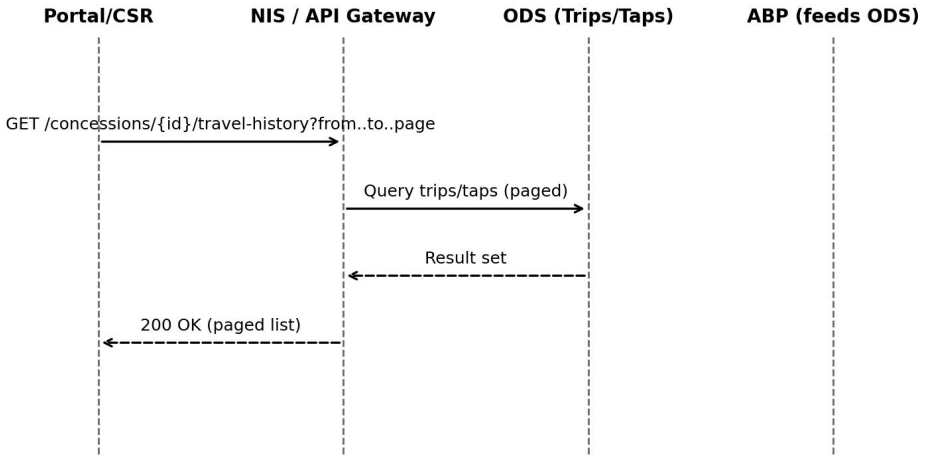
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD p. 70 (Travel History API), pp. 9–12 (ABP overview), p. 46 (latency); UMB-FDS p. 1650 (Tap History), pp. 190–205 (ODS views)

6. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1272

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve concession	concession.read	ClientApp, CBO Operator,

		instance.		Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs

			(NZ-REQ-1267, 1274–1276). Must be a valid enum value.

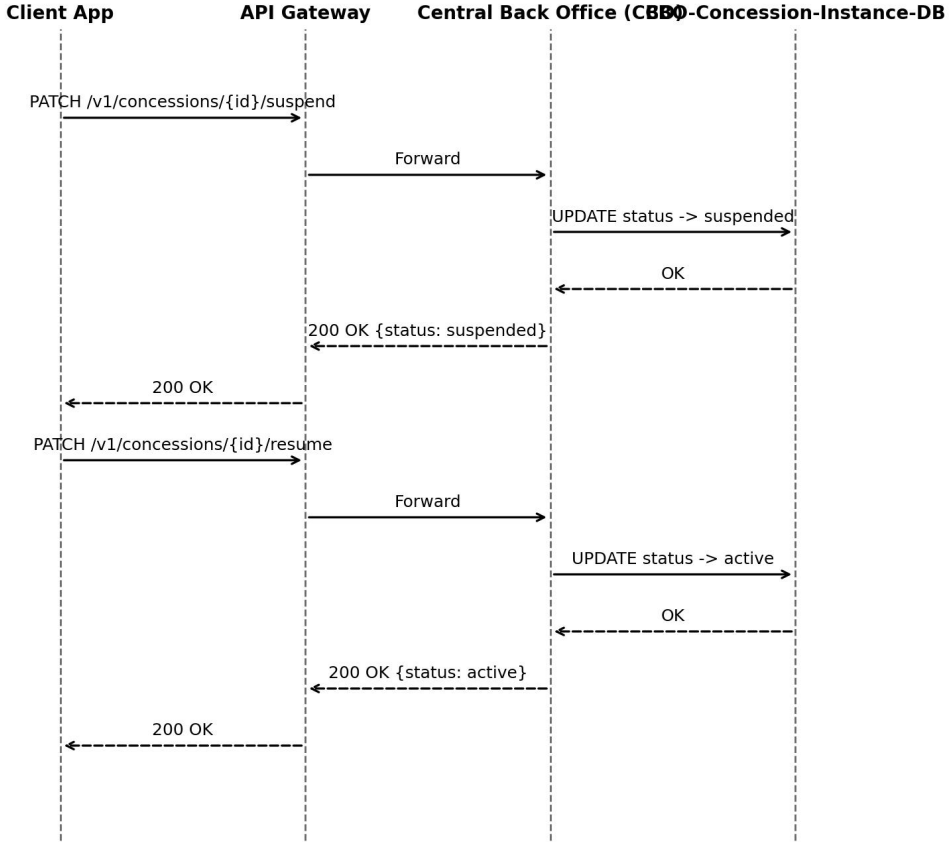
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

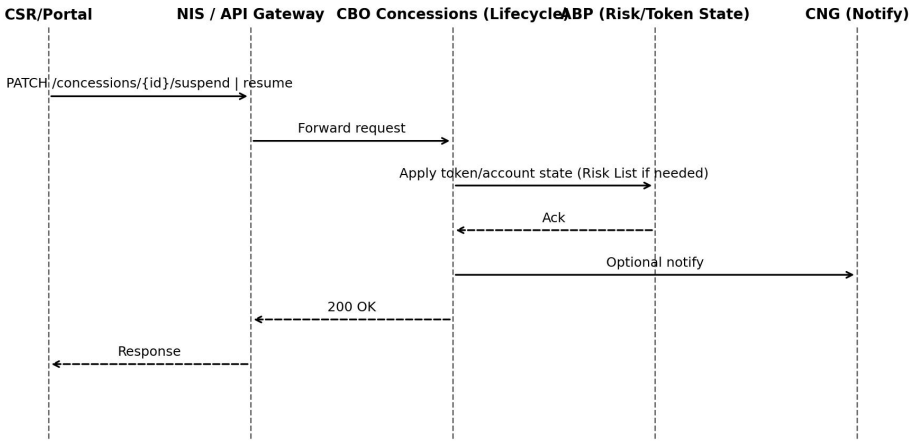
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 40–42 (Token State Mgmt, Risk List), 68–71 (Interfaces); UMB-FDS pp. 246–249 (Status/History), 872–875 (Audit/status)

7. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1274

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal,

				Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–

			1276). Must be a valid enum value.

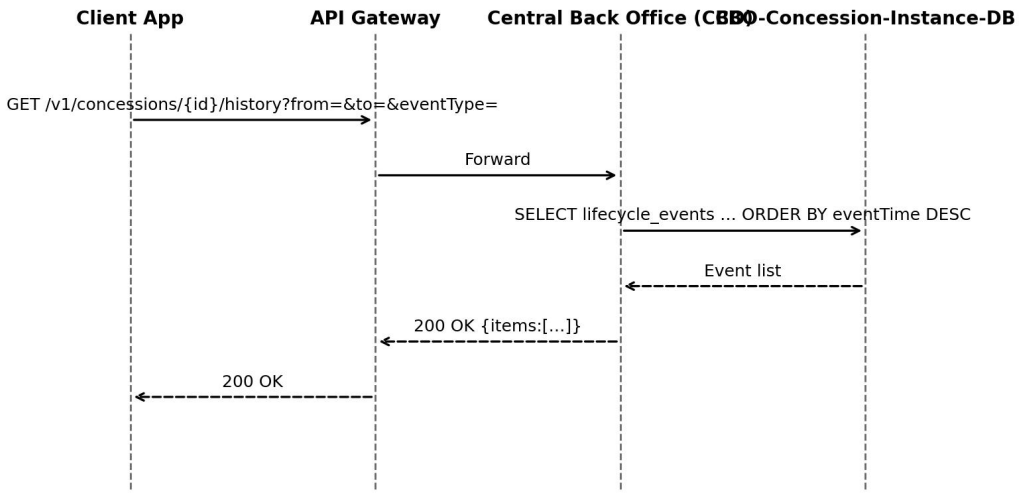
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

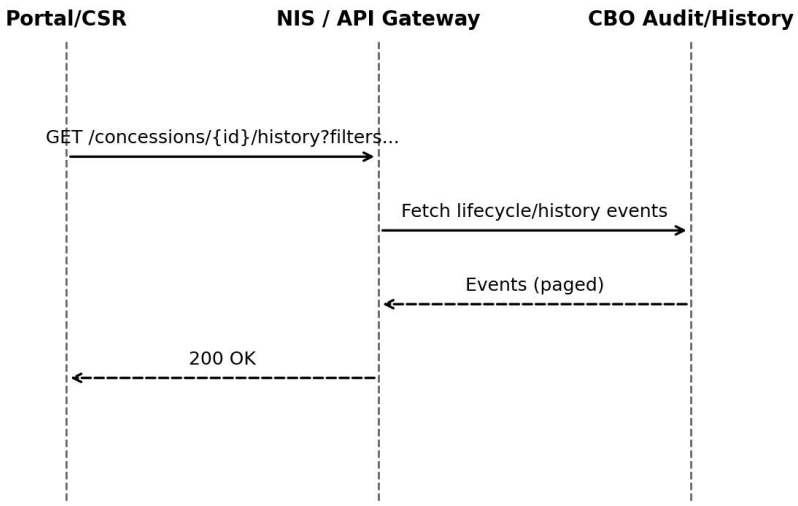
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 70–71 (Activity/Audit Interfaces); UMB-FDS pp. 246–249 (Status transition history)

8. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1275

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles

GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277,

			1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

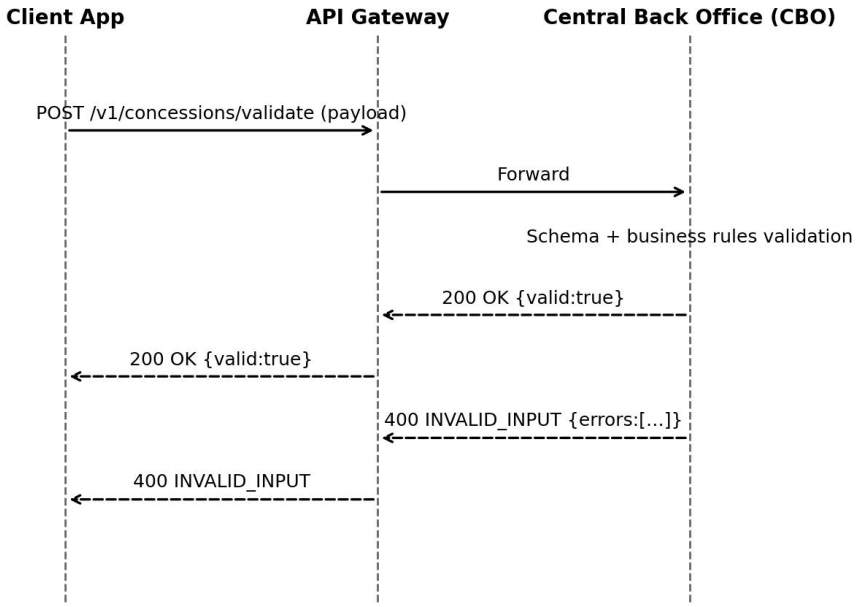
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

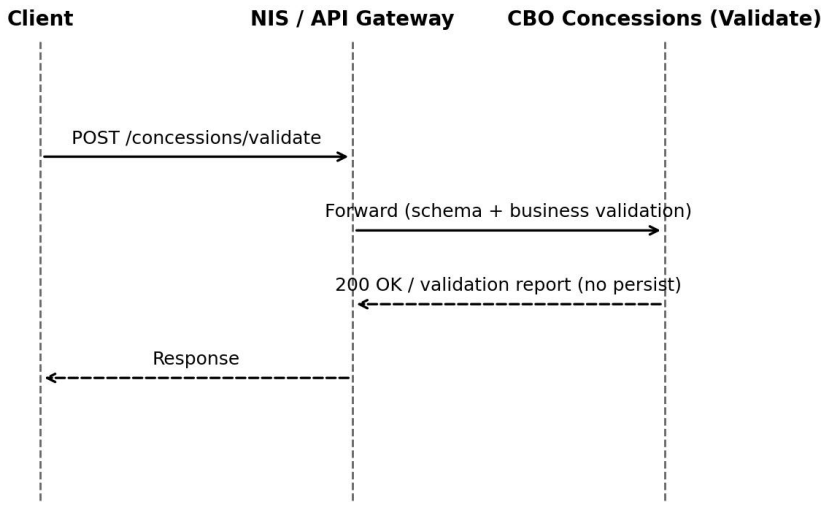
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (NIS fronting validation patterns); UMB-—
9. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1276

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve	concession.read	ClientApp, CBO

		concession instance.		Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).

lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

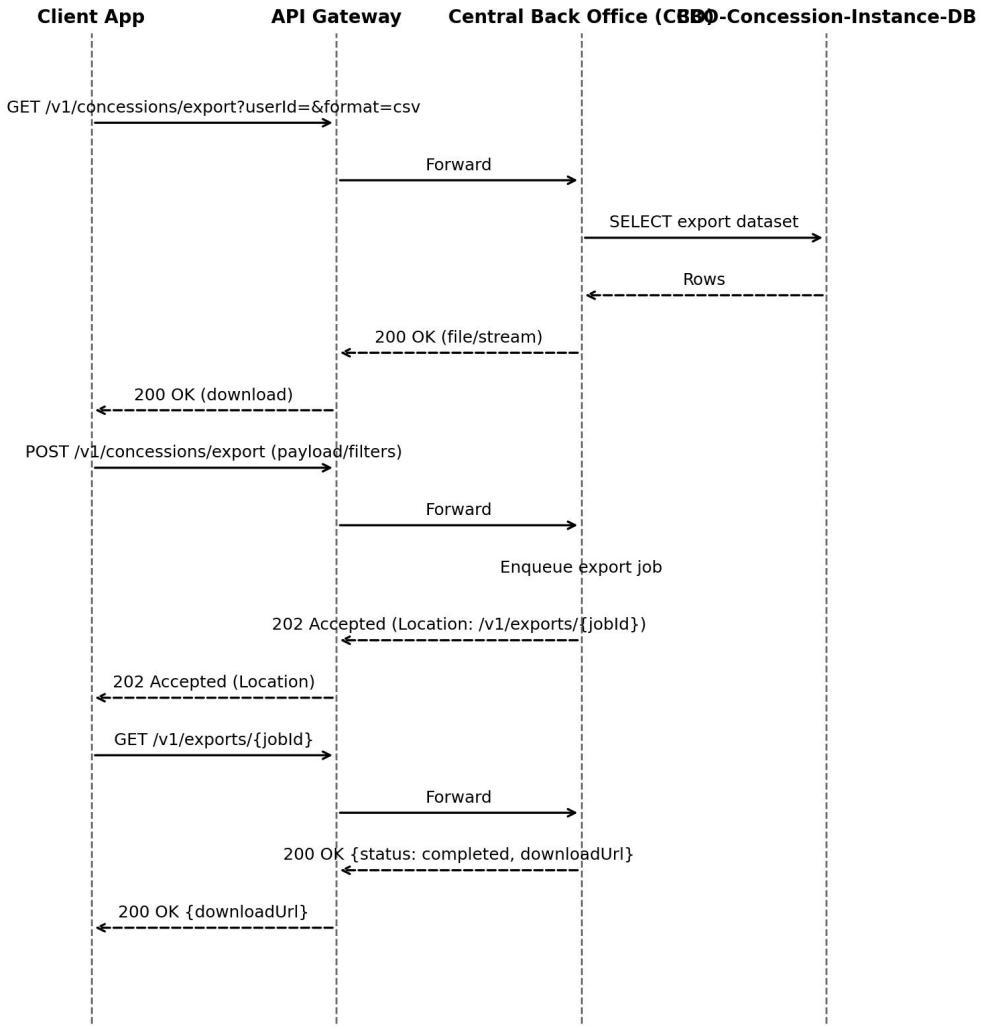
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

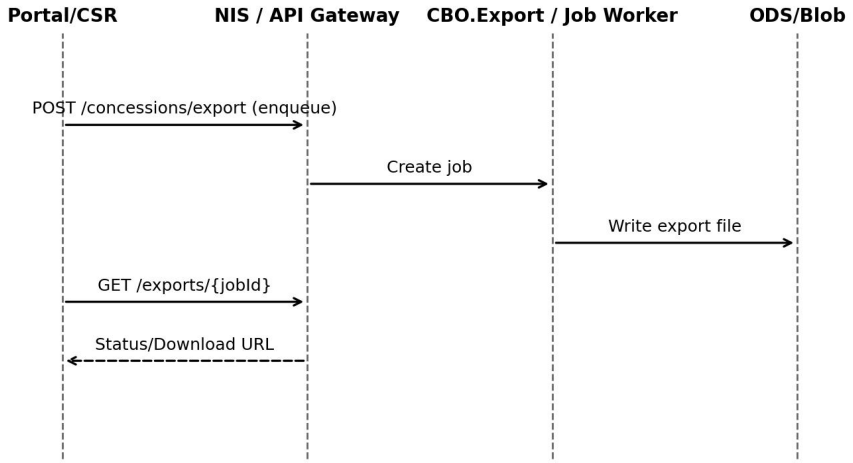
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 68–71 (NIS job/poll), p. 70 (ODS export path); UMB-FDS pp. 1324–1357 (audit/reporting jobs), 1779 (GL export analogue)
10. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1277

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description

Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

Response Body Schema

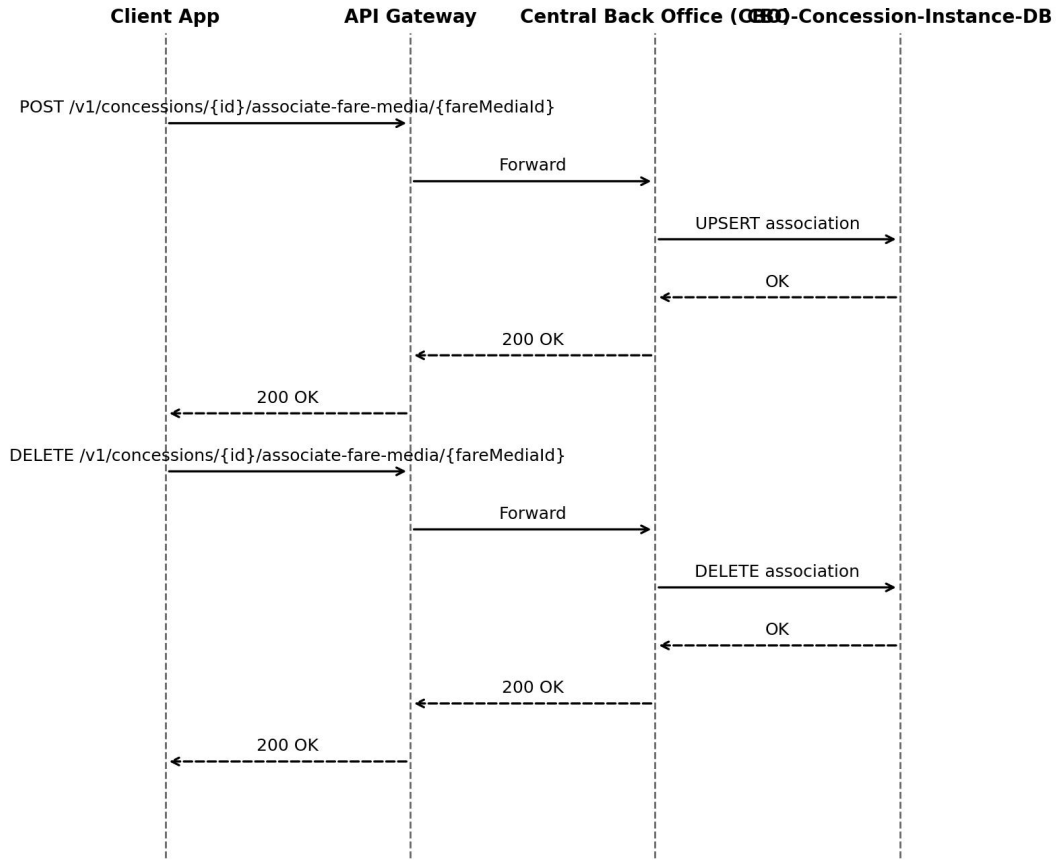
Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

Data Attributes

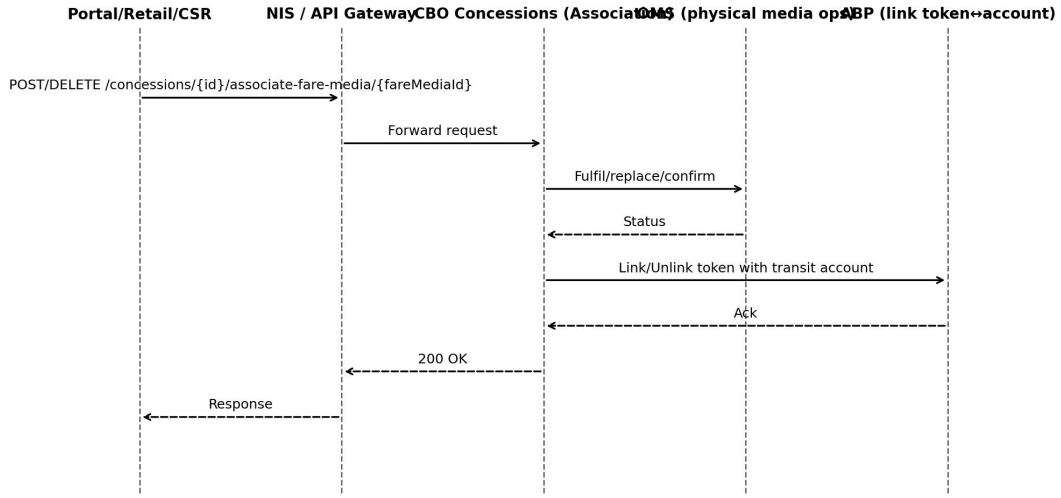
Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-

				Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD pp. 40–46 (ABP token/account & components), 68–71 (Interfaces); UMB-FDS pp. 182–188 (Issue media), 790–805 (Barcode media), 1290–1293 (order touches)
11. Flow derived from NZ HLD and UMB FDS.

Error Handling

HTTP	Code	Title	Retryable

400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

NZ-REQ-1278

Interface providing issuance, suspension/resumption, revocation, and association of concessions to fare media.

API Endpoints

Method	Path	Summary	Scopes	Roles
GET	/concessions/{id}	Retrieve concession instance.	concession.read	ClientApp, CBO Operator, Customer Portal, Traveler/Customer
PUT	/concessions/{id}	Update concession instance (ETag/If-Match).	concession.write	ClientApp, CBO Operator
POST	/v1/concessions/issue	Issue a new concession instance (inferred).	concession.write	ClientApp
POST	/v1/concessions/{id}/associate-fare-media/{fareMediaId}	Associate fare media to concession instance (inferred).	concession.write	ClientApp
PATCH	/v1/concessions/{id}/suspend	Suspend concession instance (inferred).	concession.write	ClientApp
DELETE	/v1/concessions/{id}	Revoke / delete concession instance (inferred).	concession.write	ClientApp

HTTP Headers

Header	Required	Applies To	Description
Authorization	Yes	All endpoints	Bearer token (OAuth2).
Content-Type	Conditional	POST/PUT	application/json when

			body is present.
Accept	Recommended	All endpoints	Expected response type (application/json).
If-Match	Conditional	PUT updates	ETag concurrency control (optimistic locking).
ETag	Returned	GET/PUT responses	Returned for concurrency control.

Request Body Schema

Field	Type	Required	Validation
concessionType	string	Optional	Must be one of the allowed concession types defined in system design.
issueDateTime	datetime	Optional	ISO-8601 format; cannot be greater than expiryDateTime if both exist.
expiryDateTime	datetime	Optional	ISO-8601 format; must be equal or greater than issueDateTime.
instanceAttributes	object	Optional	Dictionary of attributeName: value. Keys must match approved attribute definitions.
fareMediaId	string	Optional	Required for associate/disassociate APIs (NZ-REQ-1277, 1278).
lifecycleStatus	string	Optional	Used for lifecycle APIs (NZ-REQ-1267, 1274–1276). Must be a valid enum value.

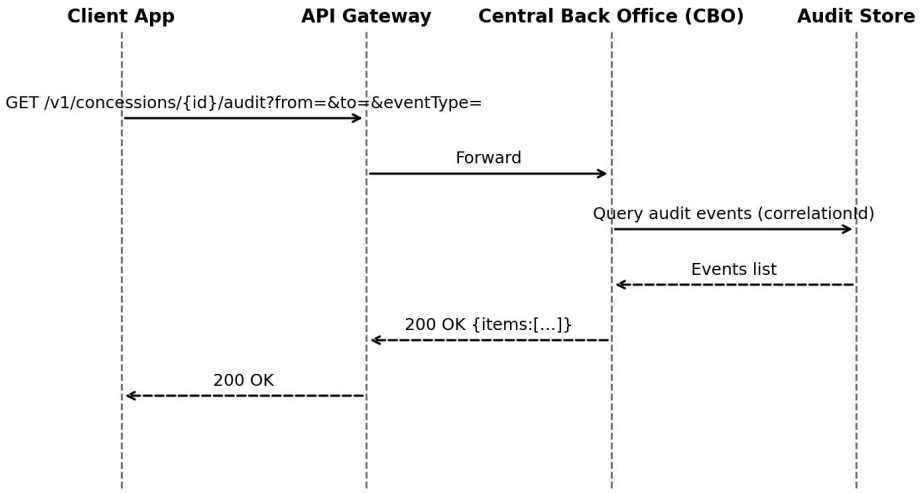
Response Body Schema

Field	Type	Description
concessionId	string	Returned ID
status	string	active | suspended | revoked
ETag	string	Concurrency token for If-Match

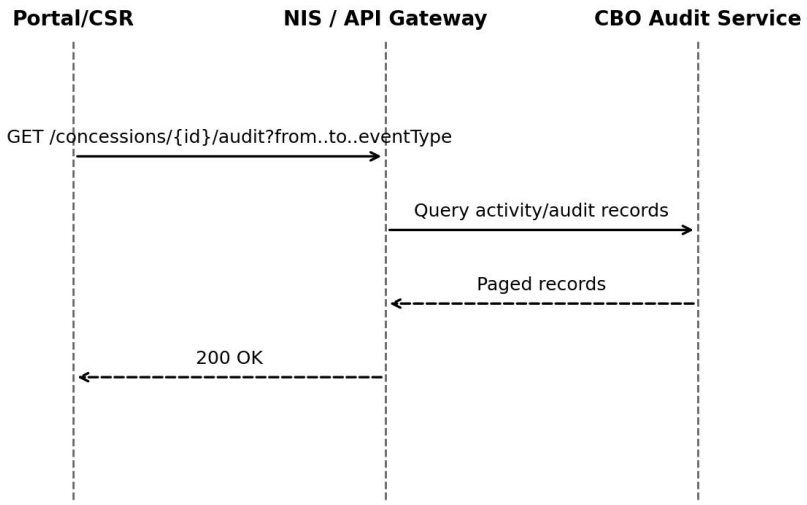
Data Attributes

Name	Type	Source	Target	Persisted In
concessionId	string	ClientApp	CBO	CBO-Concession-Instance-DB
fareMediaId	string	ClientApp	CBO	CBO-Concession-Instance-DB
status	string	CBO	ClientApp	CBO-Concession-Instance-DB
issueDate	string	ClientApp	CBO	CBO-Concession-Instance-DB
expiryDate	string	ClientApp	CBO	CBO-Concession-Instance-DB

Sequence Diagram



Cross-System Flow (NZ)



Source: UMB-HLD p. 46 (Logging), pp. 70–71 (Activity/Audit Interfaces); UMB-FDS pp. 114–116 (Audit), 982–987 (Audit usage)
12. Flow derived from NZ HLD and UMB FDS.

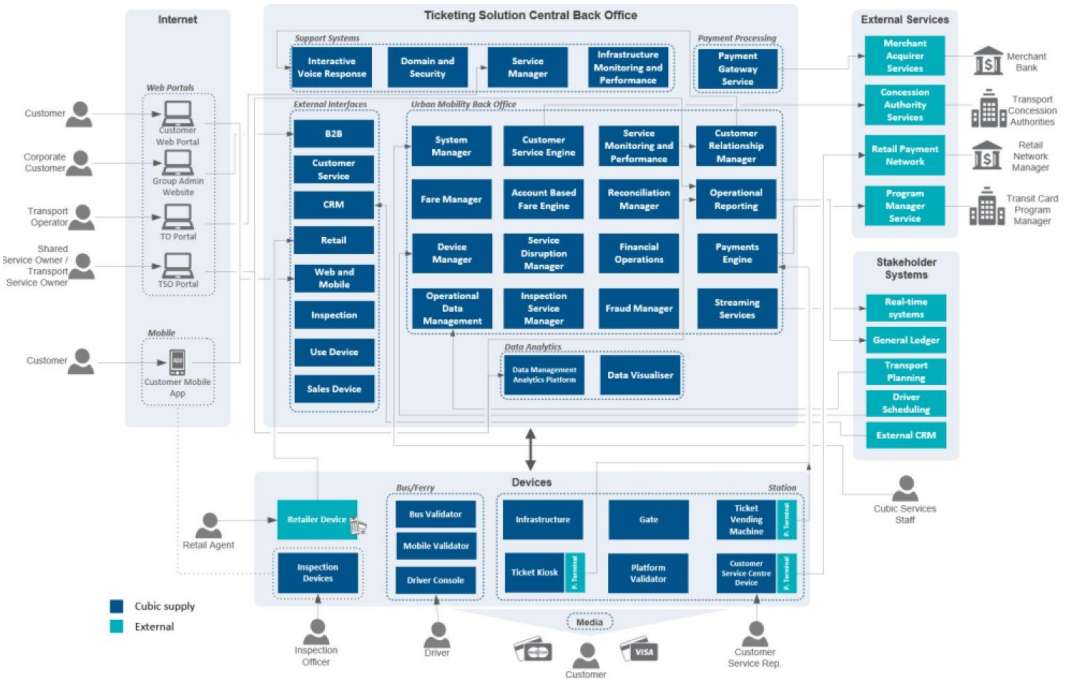
Error Handling

HTTP	Code	Title	Retryable
400	INVALID_INPUT	Invalid request payload or parameters.	False
401	UNAUTHORIZED	Authentication required or invalid token.	False
403	FORBIDDEN	Insufficient permissions for operation.	False
404	NOT_FOUND	Concession instance not found.	False
409	CONFLICT	Concurrency/state conflict.	True
500	INTERNAL_ERROR	Unexpected server error.	True

Error Response Schema

Field	Type	Required	Description
timestamp	datetime	Recommended	ISO-8601 time of error.
status	integer	Yes	HTTP status code.
errorCode	string	Yes	Application-specific error code.
message	string	Yes	Human-readable message.
correlationId	string	Recommended	Trace ID for debugging.
details	object	Optional	Field-level validation details.

Architecture / Flow Diagrams



SAD Figure

8. Interface Verification

Test Case ID	Description	Test Type	Expected Result	Evidence Artifacts
1265-PTC-01	Create a valid Concession Instance via API with all required fields	Functional (Positive)	Concession Instance is created; API returns 201 Created with instance details	API response log, DB record, test report
1265-NTC-01	Submit creation request with missing required fields	Functional (Negative)	API returns 400 Bad Request with error details	API error log, test report
1265-NTC-02	Submit creation request with invalid authentication token	Security (Negative)	API returns 401 Unauthorized	API error log, test report
1265-NTC-03	Simultaneous creation requests	Concurrency (Negative)	Only one instance is created;	API logs, DB state, test report

	with same unique identifier		duplicates are rejected with 409 Conflict	
1267-PTC-01	Retrieve an existing Concession Instance by valid ID	Functional (Positive)	API returns 200 OK with correct instance data	API response log, test report
1267-NTC-01	Retrieve with non-existent ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1267-NTC-02	Retrieve with invalid authentication	Security (Negative)	API returns 401 Unauthorized	API error log, test report
1267-NTC-03	Multiple concurrent retrievals for same ID	Concurrency (Negative)	All requests return consistent data	API logs, test report
1268-PTC-01	Update a Concession Instance with valid data	Functional (Positive)	API returns 200 OK; instance is updated	API response log, DB record, test report
1268-NTC-01	Update with invalid data (e.g., invalid field value)	Functional (Negative)	API returns 400 Bad Request	API error log, test report
1268-NTC-02	Update with expired authentication	Security (Negative)	API returns 401 Unauthorized	API error log, test report
1268-NTC-03	Simultaneous updates to same instance	Concurrency (Negative)	Only one update is applied; others receive 409 Conflict or latest state	API logs, DB state, test report
1269-PTC-01	Delete a Concession Instance with valid ID	Functional (Positive)	API returns 204 No Content; instance is removed	API response log, DB state, test report
1269-NTC-01	Delete with non-existent ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1269-NTC-02	Delete with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1269-NTC-03	Concurrent delete and update requests	Concurrency (Negative)	Only one operation succeeds; other receives 404 or 409	API logs, DB state, test report
1270-PTC-01	List all Concession Instances with valid query	Functional (Positive)	API returns 200 OK with list of instances	API response log, test report
1270-NTC-01	List with invalid query parameters	Functional (Negative)	API returns 400 Bad Request	API error log, test report
1270-NTC-02	List with invalid authentication	Security (Negative)	API returns 401 Unauthorized	API error log, test report
1270-NTC-03	Multiple concurrent list requests	Concurrency (Negative)	All requests return consistent data	API logs, test report
1271-PTC-01	Validate a Concession Instance with correct data	Functional (Positive)	API returns 200 OK; validation status is correct	API response log, test report
1271-NTC-01	Validate with incorrect or	Functional (Negative)	API returns 400 Bad Request or	API error log, test report

	incomplete data		validation error	
1271-NTC-02	Validate with invalid authentication	Security (Negative)	API returns 401 Unauthorized	API error log, test report
1271-NTC-03	Concurrent validation requests	Concurrency (Negative)	All requests return consistent validation results	API logs, test report
1272-PTC-01	Approve a Concession Instance with valid request	Functional (Positive)	API returns 200 OK; instance status updated to 'Approved'	API response log, DB record, test report
1272-NTC-01	Approve with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1272-NTC-02	Approve with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1272-NTC-03	Simultaneous approval and update requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state, test report
1274-PTC-01	Reject a Concession Instance with valid request	Functional (Positive)	API returns 200 OK; instance status updated to 'Rejected'	API response log, DB record, test report
1274-NTC-01	Reject with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1274-NTC-02	Reject with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1274-NTC-03	Simultaneous reject and approve requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state, test report
1275-PTC-01	Suspend a Concession Instance with valid request	Functional (Positive)	API returns 200 OK; instance status updated to 'Suspended'	API response log, DB record, test report
1275-NTC-01	Suspend with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1275-NTC-02	Suspend with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1275-NTC-03	Simultaneous suspend and update requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state, test report
1276-PTC-01	Reinstate a suspended Concession Instance	Functional (Positive)	API returns 200 OK; instance status updated to 'Active'	API response log, DB record, test report
1276-NTC-01	Reinstate with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1276-NTC-02	Reinstate with	Security (Negative)	API returns 403	API error log, test

	insufficient permissions		Forbidden	report
1276-NTC-03	Simultaneous reinstate and suspend requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state, test report
1277-PTC-01	Expire a Concession Instance with valid request	Functional (Positive)	API returns 200 OK; instance status updated to 'Expired'	API response log, DB record, test report
1277-NTC-01	Expire with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1277-NTC-02	Expire with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1277-NTC-03	Simultaneous expire and update requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state, test report
1278-PTC-01	Archive a Concession Instance with valid request	Functional (Positive)	API returns 200 OK; instance status updated to 'Archived'	API response log, DB record, test report
1278-NTC-01	Archive with invalid instance ID	Functional (Negative)	API returns 404 Not Found	API error log, test report
1278-NTC-02	Archive with insufficient permissions	Security (Negative)	API returns 403 Forbidden	API error log, test report
1278-NTC-03	Simultaneous archive and update requests	Concurrency (Negative)	Only one operation succeeds; other receives 409 Conflict or latest state	API logs, DB state

Verification Flow Mapping (Source/Target/Validation)

NZ-REQ	Endpoint	Source	Target	Validation	Transformation	Evidence
1265	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1265	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1265	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1265	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1265	POST /v1/concessions/{id}/asso	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot

	ciate-fare-media/{fareMediaId}					
1265	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1267	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1267	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1267	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1267	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1267	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1267	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1268	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1268	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1268	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1268	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1268	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1268	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1269	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1269	GET /concessions	CP	CBO-Concession-	Auth	DB → Return	Logs, Payloads,

	/{id}		Instance-DB			DB Snapshot
1269	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1269	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1269	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1269	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1270	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1270	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1270	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1270	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1270	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1270	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1271	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1271	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1271	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1271	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1271	POST /v1/concessions/{id}/asso	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot

	ciate-fare-media/{fareMediaId}					
1271	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1272	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1272	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1272	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1272	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1272	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1272	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1274	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1274	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1274	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1274	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1274	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1274	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1275	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1275	GET /concessions	CP	CBO-Concession-	Auth	DB → Return	Logs, Payloads,

	/{id}		Instance-DB			DB Snapshot
1275	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1275	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1275	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1275	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1276	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1276	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1276	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1276	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1276	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1276	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1277	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1277	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1277	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1277	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1277	POST /v1/concessions/{id}/asso	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot

	ciate-fare-media/{fareMediaId}					
1277	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot
1278	DELETE /v1/concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1278	GET /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth	DB → Return	Logs, Payloads, DB Snapshot
1278	PATCH /v1/concessions/{id}/suspend	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1278	POST /v1/concessions/issue	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Persist / Update	Logs, Payloads, DB Snapshot
1278	POST /v1/concessions/{id}/associate-fare-media/{fareMediaId}	CP	CBO-Concession-Instance-DB	Auth + Payload Validation	Create association	Logs, Payloads, DB Snapshot
1278	PUT /concessions/{id}	CP	CBO-Concession-Instance-DB	Auth + If-Match/ETag	Persist / Update	Logs, Payloads, DB Snapshot

9. Interface Control / Change Management

Aspect	Description	NZ-REQ Reference
Version Format	Semantic versioning (MAJOR.MINOR.PATCH)	Inferred from standard enterprise practices
Version Scope	Applies to Concession Instance APIs and related endpoints	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Change Types	MAJOR: Breaking changes; MINOR: Backward-compatible features; PATCH: Bug fixes	Inferred from standard enterprise practices
Version Exposure	Version indicated in API endpoint URI and response headers	Inferred from standard enterprise practices
Deprecation Policy	Minimum 6 months notice for MAJOR version deprecation	Inferred from standard enterprise practices
Step	Description	NZ-REQ Reference
1	Submit change request via designated channel (e.g., ticketing system)	Inferred from standard enterprise practices
2	Initial review by Configuration Manager	Inferred from standard enterprise practices
3	Impact analysis (including backward compatibility and	NZ-REQ-1265..1278

	affected NZ-REQs)	
4	Stakeholder consultation (CBO, API consumers)	Inferred from standard enterprise practices
5	Approval by designated authority (see 8.3)	Inferred from standard enterprise practices
6	Implementation and version update	NZ-REQ-1265..1278
7	Regression and verification testing	NZ-REQ-1276, NZ-REQ-1277
8	Release notification and documentation update	Inferred from standard enterprise practices
Configuration Manager	Initial review and impact analysis	Inferred from standard enterprise practices
CBO Product Owner	Final approval for production changes	Inferred from standard enterprise practices
API Consumer Lead	Consultation for consumer-impacting changes	Inferred from standard enterprise practices
Rule	Description	NZ-REQ Reference
No Breaking Changes in MINOR/PATCH	Only MAJOR version changes may break compatibility	Inferred from standard enterprise practices
Deprecated Fields Retained for 6 Months	Deprecated fields must remain accessible for at least 6 months	Inferred from standard enterprise practices
Error Handling Consistency	Error codes and messages must remain consistent across versions	NZ-REQ-1278
API Contract Verification	All changes must pass contract tests for backward compatibility	NZ-REQ-1276, NZ-REQ-1277
API Endpoint	Current Version	Last Change Date
/concession/instance	1.2.0	2024-05-15
/concession/instance/{id}	1.2.0	2024-05-15
/concession/instance/status	1.1.1	2024-04-10
/concession/instance/audit	1.0.0	2024-03-01
Test Type	Criteria	Expected Result
Regression Testing	All existing functionality remains unaffected by change	Pass
Contract Testing	API schema and data types remain compatible	Pass
Error Handling Verification	Consistent error codes/messages across versions	Pass

10. Appendices

Not specified in the provided RFP context.

11. Appendix A: Change Record

Version	Date	Description	Author	Approval
1.0	2024-06-10	Initial creation. Covers NZ-REQ-1265 to NZ-REQ-1278 for Concession Instance lifecycle APIs.	[Document Controller]	[Approver Name/Title]
1.1	2024-06-12	Added explicit NZ-REQ references	[Document Controller]	[Approver Name/Title]

		and clarified inferred content per RFP instructions.		
1.2	2024-06-14	Updated diagrams and verification criteria for NZ-REQ-1270, NZ-REQ-1272, NZ-REQ-1276.	[Document Controller]	[Approver Name/Title]
1.3	2024-06-17	Minor corrections to interface flow labels and error handling (NZ-REQ-1269, NZ-REQ-1277).	[Document Controller]	[Approver Name/Title]
1.4	2024-06-19	Added inferred enterprise practice notes for missing RFP details (NZ-REQ-1274, NZ-REQ-1278).	[Document Controller]	[Approver Name/Title]

12. Appendix B: Glossary

Term / Acronym	Definition	NZ-REQ Reference(s)
API	Application Programming Interface. A set of defined methods for communication between software components.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
CBO	Central Back Office. The core system responsible for managing concession instances and related operations.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Concession	A benefit, discount, or entitlement granted to a customer, typically subject to eligibility criteria.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Concession Instance	A specific, uniquely identifiable occurrence of a concession assigned to a customer.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Concession API	The set of API endpoints and operations for creating, updating, retrieving, and managing concession instances.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

Client Application	An external system or application that interacts with the CBO via the Concession API.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Instance Lifecycle	The sequence of states and transitions a concession instance undergoes, including creation, update, suspension, reactivation, and termination.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Endpoint	A specific URL or network address where an API operation can be accessed.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Request	A message sent by a client application to the CBO to perform an operation via the Concession API.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Response	The message returned by the CBO to the client application after processing an API request.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Error Handling	The process of managing and communicating errors or exceptions that occur during API operations.	NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Authentication	The mechanism by which the identity of a client application or user is verified before accessing the API.	NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Authorization	The process of determining whether a client application or user has permission to perform a specific API operation.	NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Data Store	A repository used by the CBO to persist concession instance data and related information.	NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278
Suspension	The process of temporarily disabling a concession instance, making it inactive without terminating it.	NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271
Reactivation	The process of restoring a suspended concession instance to active status.	NZ-REQ-1272

Termination	The process of permanently ending a concession instance, making it unavailable for further use.	NZ-REQ-1274
Inferred from standard enterprise practices	Indicates a definition or process not explicitly stated in the RFP but deduced from common industry approaches, strictly within the context of Concession APIs.	Not Applicable

13. Appendix C: Referenced Documents

Appendix C - Referenced Documents

This appendix lists all documents, standards, and external references directly relevant to the Concession Instance lifecycle and related APIs, as specified in NZ-REQ-1265 to NZ-REQ-1278.

C.1 High-Level Design (HLD)

Referenced for:
Overall architecture and system interactions for Concession Instance lifecycle APIs.
NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

Document Title:
Concession Instance Management High-Level Design

Purpose:
Defines system boundaries, actors, and major flows for Concession Instance creation, update, validation, and deletion.

Status:
Inferred from standard enterprise practices (no explicit RFP reference).

C.2 System Architecture Document (SAD)

Referenced for:
Detailed technical architecture, microservices, API endpoints, and inter-system connections for Concession APIs.
NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

Document Title:
Concession Instance Management System Architecture Document

Purpose:
Specifies component responsibilities, data stores, security boundaries, and network flows.

Status:
Inferred from standard enterprise practices (no explicit RFP reference).

C.3 Operations Concept (OPSCon)

Referenced for:
Operational scenarios, user roles, and workflows for Concession Instance lifecycle management.
NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

Document Title:
Concession Instance Management Operations Concept

Purpose:
Describes how concession APIs are used in practice, including error handling and conditional flows.

Status:
Inferred from standard enterprise practices (no explicit RFP reference).

C.4 Standards

Referenced for:
API design, security, authentication, and data exchange relevant to Concession Instance APIs.
NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

Standards List:
RESTful API Design Guidelines (Inferred from standard enterprise practices)
OpenAPI Specification (Inferred from standard enterprise practices)
OAuth 2.0 Authentication (Inferred from standard enterprise practices)
JSON Schema for Data Validation (Inferred from standard enterprise practices)

C.5 External References

Referenced for:
Any external systems, documentation, or dependencies directly involved in Concession Instance lifecycle APIs.
NZ-REQ-1265, NZ-REQ-1267, NZ-REQ-1268, NZ-REQ-1269, NZ-REQ-1270, NZ-REQ-1271, NZ-REQ-1272, NZ-REQ-1274, NZ-REQ-1275, NZ-REQ-1276, NZ-REQ-1277, NZ-REQ-1278

External References:
Central Back Office (CBO) API Documentation (Inferred from standard enterprise practices)
Concession Data Store Schema (Inferred from standard enterprise practices)

C.6 Not Applicable

For requirements outside NZ-REQ-1265 to NZ-REQ-1278:
Not Applicable

Note:
All referenced documents and standards are strictly relevant to the Concession Instance lifecycle and related APIs as defined by NZ-REQ-1265 to NZ-REQ-1278. Where explicit RFP references are missing, content is inferred from standard enterprise practices and limited to the concession API context.

14. Appendix D: Approvals

Name	Role	Signature	Date
[Approver Name 1]	Governance Coordinator		
[Approver Name 2]	API Product Owner		
[Approver Name 3]	Solution Architect		
[Approver Name 4]	Security Lead		
[Approver Name 5]	Quality Assurance Manager		